SMART HOME AUTOMATION SYSTEM

Short Project Report
Cognevance Technologies – Embedded Systems & IoT Internship

Project Type: Software/Simulation-Based IoT Project

Simulation Platform: Wokwi

Dashboard: HTML, CSS, JavaScript and Chart.js

1. Project Overview

The Smart Home Automation System is a simulation-based IoT project developed to monitor environmental conditions and demonstrate automatic appliance control. The implemented system uses a DHT22 sensor for temperature and humidity, an LDR module for light intensity, a PIR sensor for motion detection, a relay module for appliance control, and LEDs for status indication. Sensor information is presented through a web dashboard with status indicators, alerts and historical graphs.

2. Objectives

• Monitor temperature and humidity.
• Detect ambient light level and motion.
• Apply threshold-based automation for fan and room-light control.
• Display sensor information through a web dashboard.
• Provide alerts for abnormal conditions.
• Demonstrate communication between structured sensor data and the dashboard.
• Test the system under normal and abnormal conditions.

3. System Components

Component	Function
Arduino Uno	Main controller for sensor processing and control logic
DHT22	Temperature and humidity measurement
LDR module	Ambient light intensity measurement
PIR sensor	Motion detection
Relay module	Switching/control of an appliance
Red / Green LEDs	Alert and status indication
Web dashboard	Monitoring, graphs, alerts and appliance status

4. System Implementation

The embedded system was assembled and tested in Wokwi. The DHT22 provides temperature and humidity readings, while the LDR and PIR modules provide light and motion information. The controller evaluates the sensor conditions and drives the relay and status LEDs. A software communication layer represents the sensor state in JSON format, which is read by the web dashboard. The dashboard updates the sensor cards, appliance status, alerts and graphs periodically.

5. Wokwi Circuit and Serial Output

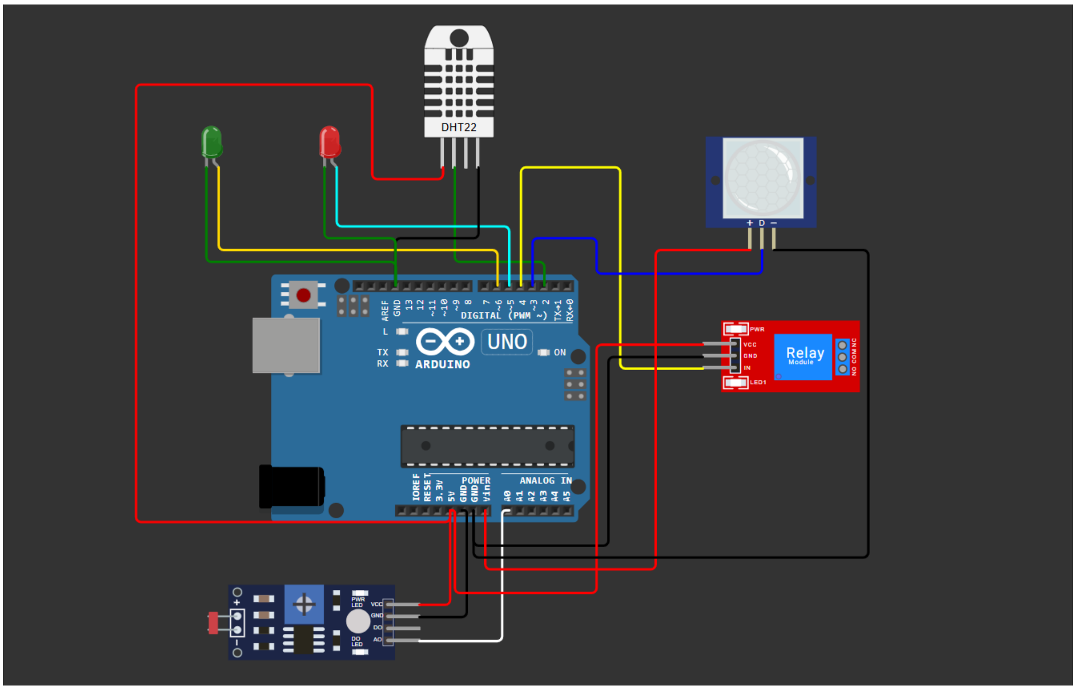

Figure 1. Wokwi simulation circuit used for the Smart Home Automation System.

Figure 2. Serial Monitor showing temperature, humidity, light, motion and control status.

6. Web Dashboard

The web dashboard provides a visual interface for monitoring the system. It displays temperature, humidity, light condition, motion status, automation mode, connection status, fan/light state, alerts and historical sensor graphs.

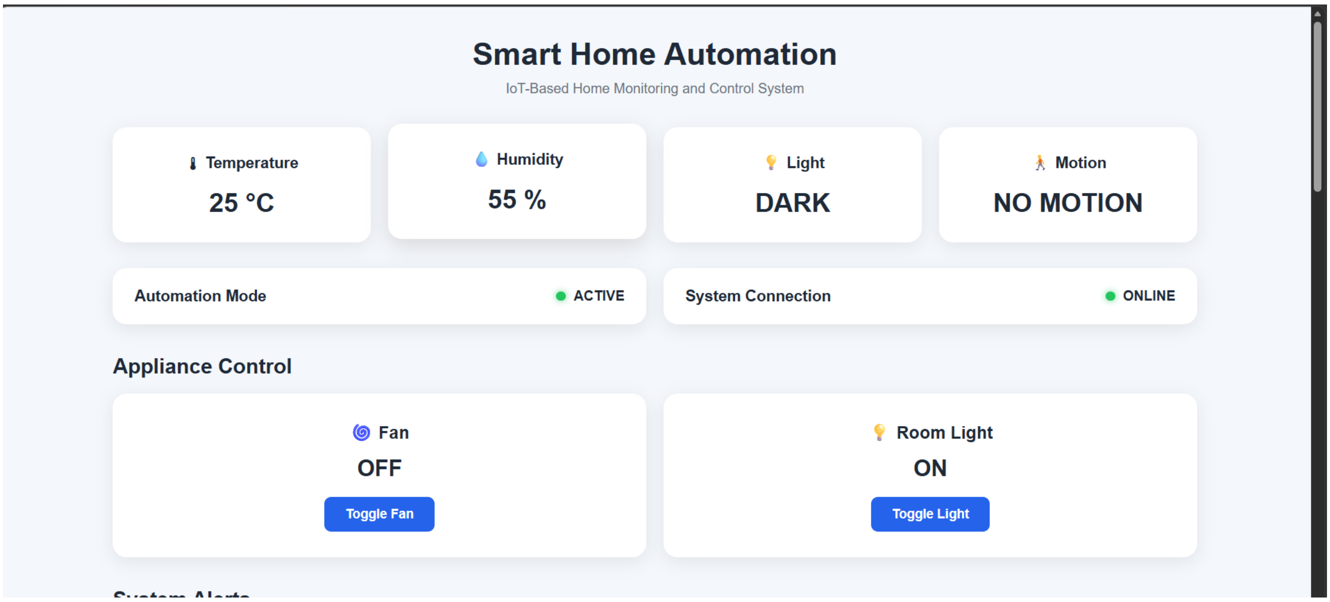

Figure 3. Smart Home Automation monitoring and appliance-control dashboard.

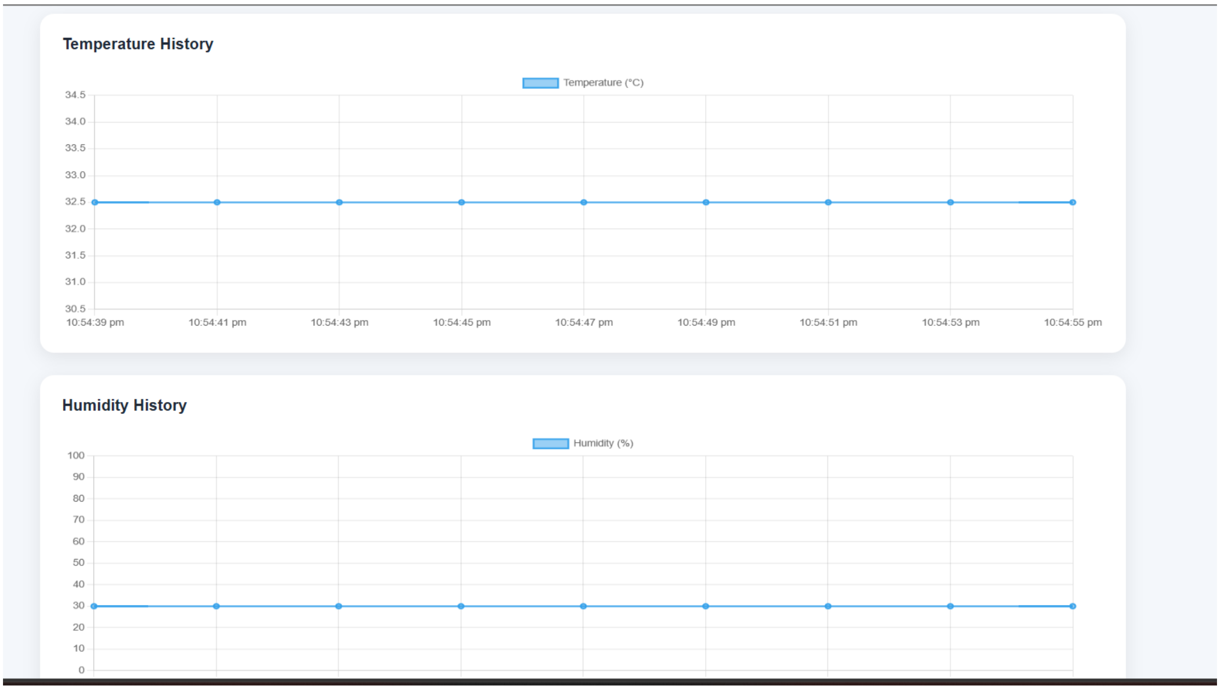

Figure 4. Temperature and humidity history graphs generated by the dashboard.

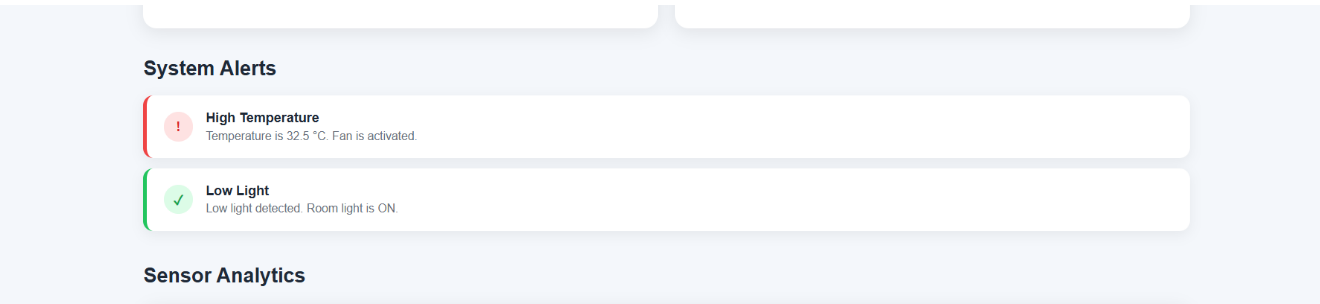

Figure 5. Example dashboard alert for high temperature and low light conditions.

7. Automation Logic

Temperature: When the temperature reaches the configured high-temperature threshold, the fan control is activated and a high-temperature alert is generated.
Light: When the measured light value is below the configured threshold, the environment is treated as dark and the room-light control is activated.

Motion: When the PIR sensor detects movement, the dashboard reports motion and generates a motion alert.
Humidity: When humidity reaches the configured high-humidity threshold, a high-humidity alert is generated.

8. Testing and Results

Test	Condition	Observed Result	Status
T1	Normal condition	Normal monitoring	PASS
T2	Temperature = 32.5 °C	Fan ON / high-temperature alert	PASS
T3	Motion = TRUE	Motion detected / alert	PASS
T4	Light value = 300	DARK / room light ON	PASS
T5	Humidity = 80%	High-humidity alert	PASS
T6	Communication source unavailable	OFFLINE / communication error / recovery	PASS

9. Results and Outcome

The simulation successfully demonstrates multi-sensor data acquisition, threshold-based automation, relay control, dashboard visualization, graphing, alert generation and communication-status monitoring. The six defined functional tests were completed successfully. The project therefore provides a working software/simulation demonstration of a basic smart-home monitoring and automation system.

10. Future Improvements

Future versions can replace the simulated communication layer with direct Wi-Fi/MQTT communication using an ESP32, add cloud data storage, provide mobile access, implement remote appliance control and add energy-consumption monitoring.

11. Conclusion

The Smart Home Automation System was successfully designed, simulated and tested. The project combines embedded sensing, automatic control and a web-based monitoring interface. The completed Wokwi circuit, serial output, dashboard, graphs and test results provide evidence of the implemented functionality.